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Details Questions

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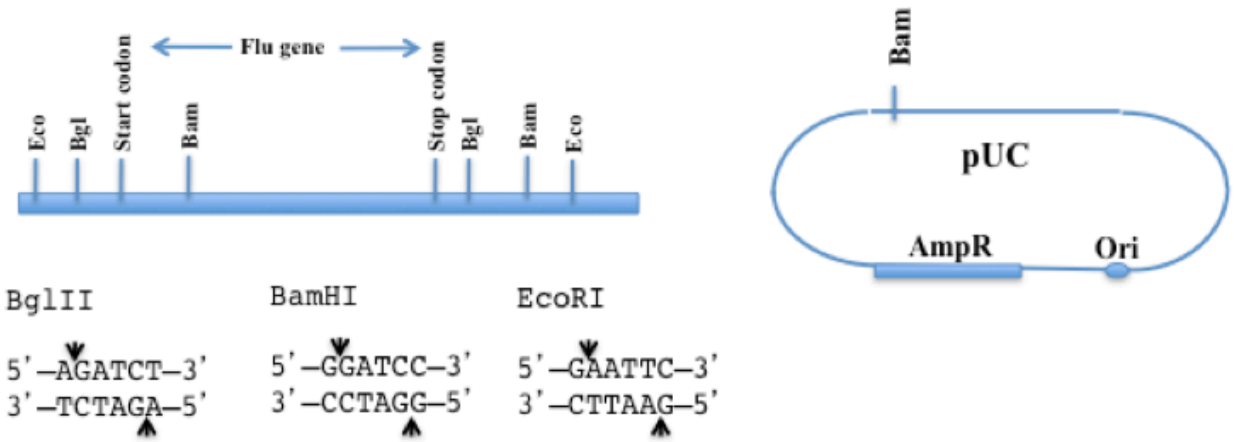
Question 0.8 pts

You seek to clone the Flu gene into the plasmid pUC. Which enzyme would you digest the Flu gene with to clone the entire gene into the unique BamHI site of pUC?



- BamHI
- EcoRI
- BglII
- All of the above
- None of the above

Question 1 pts

You isolated a gene expressed in differentiated neurons in mice. You then fused various fragments of the gene (shown as dark lines) to a GFP reporter lacking either enhancers or a promoter. The resulting clones were introduced into neurons in tissue culture, and the level of GFP expression was monitored by looking for green fluorescence. From the results that follow, which region contains the promoter? Regions are indicated by the orange sections.

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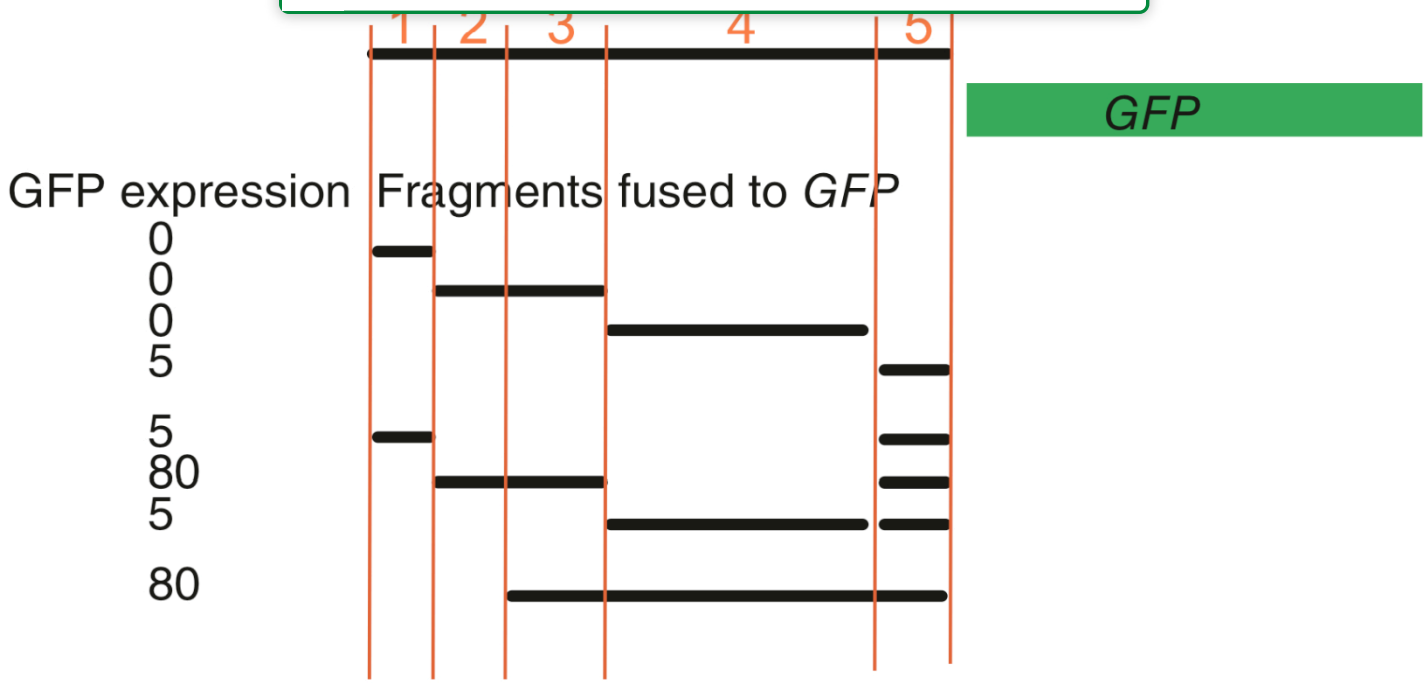

- 1
- 2
- 3
- 4
- 5

Question 0.8 pts

Which region, from above, contains the neuronal enhancer?

- 1
- 2
- 3
- 4
- 5

Question 0.9 pts

Why was the GFP reporter construct introduced into neurons in tissue culture?

What would happen if you introduced the GFP reporter into lung cells in tissue culture?

- [ ] The gene candidate fused to the reporter was differentially expressed in neurons.
- [ ] The researchers were interested in genes underlying neuron cell identities.
- [ ] No expression is likely to occur in lung cells.
- [ ] The gene will be expressed at similar levels to neurons.

Question 0.9 pts

Transgenic organisms can he

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- individuals (without a tran
- the genes that cause disease are always dominant
- the genes that cause diseases are always recessive
- strains used to create transgenic organisms carry no mutations (all their genes are wild-type)

Question 0.8 pts

Sometimes, genes transferred into the mouse genome by pronuclear injection disrupt a gene at the (random) site of integration, resulting in a mutation.

In one such case, investigators identified a recessive mutation that causes limb deformity in transgenic mice.

The insertion mutation was mapped to chromosome 2 of mice in a region where a recessive mutation called *limb deformity (ld)* had previously been identified. Mice carrying the *ld* mutation are available from a major mouse research laboratory.

How could you tell if the *ld* mutation was in the same gene as the transgene insertion mutation? I.e., what type of test would you perform?

complementation test
rescue assay
Complementation Test
Complementation test
Rescue test
rescue test
Rescue Test
Rescue Assay
Rescue assay

Question 0.8 pts

Through a GWAS analysis to identify SNPs associated with diabetes, you identify a SNP that is close to the gene CALM2.

CALM2 is part of the adenosine monophosphate-activated protein kinase signaling pathway, which is known to stimulate glucose metabolism. Because of the known function of CALM2, you decide to design an experiment to test if the SNP you identified is in an enhancer for CALM2.

The risk allele of the SNP is A and the other allele is G.

Identify the experiments that would test the whether SNP A modulates CALM2 gene expression.

- [ ] Fuse GFP to different alleles of the locus, and test their impact on CALM2 expression
- [ ] Fuse GFP to different segments of the locus, and test their impact on CALM2 expression
- [ ] Fuse GFP to different segments of CALM2, and test their impact on gene expression
- [ ] Fuse GFP to different alleles of CALM2, and test their impact on gene expression

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